

Chapter 6: LAN Security: Attacks, Detection & Defense

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Objective:

Understand common **LAN-level attacks**, their impact on **network protocols**, and learn how to **detect and prevent** them through switch security configuration in **Cisco Packet Tracer**.

Lecture Overview

1. LAN Security Overview
2. LAN Attack Vectors
3. Protocol Weaknesses (ARP, DHCP, VLAN)
4. Attack Demonstrations in Packet Tracer
5. Detection & Analysis
6. Defense: Switch Security Mechanisms
7. Troubleshooting & Best Practices

1. LAN Security Overview

LAN (Local Area Network):

An internal network interconnecting computers, switches, and routers.

Security Goal:

Ensure **confidentiality, integrity, and availability** at Layer 2.

Common Risks:

- Unauthorized access
- Internal spoofing
- Traffic interception
- Rogue DHCP or ARP poisoning
- VLAN hopping

2. LAN Attack Vectors

Attack Type	Description	Goal
MAC Flooding	Fills switch CAM table with fake MACs	Force switch to broadcast traffic
DHCP Spoofing	Fake DHCP server assigns wrong gateway/DNS	Redirect or block user traffic
ARP Spoofing	Sends fake ARP replies to impersonate gateway	Enable MITM/sniffing
VLAN Hopping	Exploits trunk misconfiguration	Access unauthorized VLANs

3. Effect on Network Protocols

Protocol	Weakness	Exploit Impact
ARP	No authentication	MITM via spoofed ARP replies
DHCP	No server validation	Rogue DHCP server poisoning
STP	Default trust of root bridge	Force topology change
802.1Q (VLAN)	Mis-tagging possible	Unauthorized VLAN access
ICMP	No session control	Network reconnaissance

4. Demonstration: LAN Attacks in Packet Tracer

Topology Overview

Device	Role	VLAN	Interface
PC0	Victim	VLAN 10	Fa0/1
PC1	Attacker	VLAN 10	Fa0/2
Server	Gateway/DHCP	VLAN 10	Fa0/24
Switch	Target device	—	—

(A) MAC Flooding Attack

Concept:

Attacker floods switch with thousands of fake MAC addresses.

Effect:

Switch CAM table overflows → broadcasts all frames → attacker can sniff all traffic.

Simulate in Packet Tracer:

- Configure PC1 as attacker.
- Send multiple fake source MACs using simulation mode.
- Observe switch flooding packets to all ports.

(A) MAC Flooding Attack

Detection:

```
show mac address-table
```

- Many unknown MACs from one port = suspicious.

Prevention:

```
interface fa0/2
  switchport port-security
  switchport port-security maximum 2
  switchport port-security violation shutdown
```


(B) DHCP Spoofing Attack

Concept:

Attacker runs a fake DHCP service assigning wrong gateway/DNS.

Effect:

Victim receives malicious network configuration → traffic redirected or blocked.

Simulate:

- Configure PC1 as DHCP server.
- Server assigns wrong gateway (e.g., 192.168.1.254).

(B) DHCP Spoofing Attack

Detection:

```
show ip dhcp binding
```

– Incorrect gateway or DNS for clients.

Prevention:

```
ip dhcp snooping  
ip dhcp snooping vlan 10  
interface fa0/24  
  ip dhcp snooping trust
```

(C) ARP Spoofing / Poisoning

Concept:

Attacker sends fake ARP replies:

"Gateway IP is at Attacker's MAC."

Effect:

Victim traffic passes through attacker (MITM).

Simulate:

- Use PC1 to send ARP replies to PC0 and gateway.
- Observe ARP table on PC0.

(C) ARP Spoofing / Poisoning

Detection:

On victim:

```
arp -a
```

Gateway MAC differs → spoof detected.

Prevention:

```
ip dhcp snooping  
ip arp inspection vlan 10  
interface fa0/24  
ip arp inspection trust
```

(D) VLAN Hopping

Concept:

Attacker crafts packets with double VLAN tags to access another VLAN.

Effect:

Cross-VLAN packet leakage (data exfiltration or sniffing).

Detection:

- Unexpected inter-VLAN communication.
- Logs showing unauthorized VLAN IDs.

(D) VLAN Hopping

Prevention:

```
interface range fa0/1 – 20  
  switchport mode access  
  switchport access vlan 10  
  switchport nonegotiate
```

Disable trunking on access ports.

5. Detecting LAN Attacks

Techniques:

- Monitor MAC table size and anomalies.
- Compare DHCP bindings and ARP tables.
- Use IDS/IPS for Layer 2 signatures.
- Enable logging and SNMP traps on switches.

Cisco Verification Commands:

```
show port-security  
show ip dhcp snooping binding  
show ip arp inspection statistics  
show log
```

6. Defense: Switch Security Mechanisms

Feature	Protects Against	Key Command
Port Security	MAC Flooding	<code>switchport port-security</code>
DHCP Snooping	DHCP Spoofing	<code>ip dhcp snooping</code>
Dynamic ARP Inspection	ARP Spoofing	<code>ip arp inspection vlan</code>
VLAN Pruning	VLAN Hopping	<code>switchport mode access</code>
BPDU Guard	STP Attacks	<code>spanning-tree bpduguard enable</code>

Configuration Summary

- Rogue DHCP blocked/Spoofed ARP dropped/MAC flooding prevented

```
! Enable Layer 2 protections
ip dhcp snooping
ip arp inspection vlan 10
!
interface fa0/1
  switchport mode access
  switchport access vlan 10
  switchport port-security
  switchport port-security maximum 2
  switchport port-security violation shutdown
!
interface fa0/24
  ip dhcp snooping trust
  ip arp inspection trust
```

7. Demonstration Flow (Packet Tracer)

Step	Activity	Expected Result
1	Enable security features	All ports secured
2	Launch MAC flood	Port shuts down after violation
3	Launch DHCP spoof	Rogue DHCP blocked
4	Send fake ARP	Packet dropped by switch
5	Check logs	Violation counters increase

7. Demonstration Flow (Packet Tracer)

Verification Commands:

```
show interfaces status err-disabled  
show port-security interface fa0/2  
show ip dhcp snooping statistics  
show ip arp inspection statistics
```

8. Troubleshooting

Symptom	Likely Cause	Resolution
Clients fail to get IP	Untrusted DHCP port	Trust uplink interface
Port down (err-disable)	Too many MACs	Increase or clear sticky MACs
DHCP blocked	VLAN not enabled	Add <code>ip dhcp snooping vlan</code>
ARP still spoofed	DAI off	Verify ARP inspection active

9. Best Practices

- Enable *port security* on all access ports.
- Use *sticky MAC* for fixed endpoints.
- Trust only uplink and trunk ports.
- Disable unused ports (shutdown).
- Monitor logs for violations.
- Combine with NAC/802.1X for enterprise-grade control.

10. Reflection Questions

1. Why does MAC flooding enable packet sniffing?
2. How does DHCP Snooping support ARP Inspection?
3. What happens when you trust a malicious port?
4. How can you visualize ARP spoofing using Packet Tracer Simulation?

End of Lecture

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